

Gopi Krishna Vicharapu

Data Analyst

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PROFESSIONAL SUMMARY

Results-driven IT professional with 4+ years of expertise in designing and deploying end-to-end Data Warehouse solutions using SSIS, Informatica, and AWS Glue. Proficient in advanced SQL, Python (Pandas, Scikit-learn), and cloud platforms (AWS, Azure) for scalable data pipelines and machine learning solutions. Skilled in creating interactive dashboards with Tableau and Power BI and automating workflows using Alteryx. Adept at optimizing database performance, building RESTful APIs, and enabling real-time data-driven decisions. Delivered actionable insights and process improvements to drive strategic growth and operational efficiency.

TECHNICAL SKILLS

Working Methodologies	SDLC, Agile, SCRUM, Kanban
Programming Language	Python, R, SQL
Visualization & BI Tools	Tableau, Power BI, Microsoft Excel (Pivot tables, VLOOKUP), SSIS, SSAS, SSRS
Database	RDBMS (MySQL, SQL Server, PostgreSQL), NoSQL (Mongo DB, Dynamo DB)
Cloud Technologies	AWS (Redshift, S3, EC2, Lambda), Azure (Data Factory, Databricks)
Data Science Libraries	NumPy, Pandas, Plotly, Matplotlib, Seaborn, ggplot2, Pyspark, TensorFlow, Scikit-learn
Big Data Engineering	Hadoop, Apache Spark, Kafka, Pig, Hive, Apache Airflow, DBT
Version control tools	Git and Github
Other Skills	Data Cleaning, Data Wrangling, Critical Thinking, Problem Solving, Presentation Skills
Operating System	MacOS, Windows, Linux

EXPERIENCE

Data Analyst | Cloud Systems – Charlotte, NC

Jan 2023 –Present

- Design and optimize real-time Tableau/Power BI dashboards to monitor cloud infrastructure, reducing system downtime by 20% and enabling data-driven decision-making.
- Automate ETL workflows using Python and AWS Glue, streamlining data integration processes and cutting integration time by 40%.
- Consolidate and unify multi-source data into a centralized Snowflake data warehouse, improving cross-team data accessibility by 35% and enhancing reporting efficiency.
- Deploy predictive machine learning models in AWS SageMaker, reducing cloud operational costs by 25% through proactive resource optimization.
- Develop SQL-based server health monitoring tools, decrease unexpected system failures by 30% and improve operational reliability.
- Integrate AWS S3 and RDS with Snowflake, enhancing query performance by 18% and enabling faster data retrieval for analytics.
- Conduct exploratory data analysis (EDA) using Pandas and Matplotlib, identifying cost-saving opportunities and driving a 15% improvement in cloud cost optimization.
- Lead end-to-end machine learning pipelines to predict server performance issues, enabling proactive scaling and reducing system failures by 20%.
- Built interactive, real-time monitoring dashboards, empowering teams to reduce system downtime by 20% and improve operational efficiency.

Key Achievements:

- Reduced integration time by 40% through automation.
- Enhanced forecasting accuracy and optimized resource usage with ML models.
- Built real-time monitoring dashboards, reducing system downtime by 20%.

Data Analyst | Bytecraft System, Hyderabad-India

Aug 2020 - Dec 2022

- Designed and optimized real-time Tableau/Power BI dashboards to monitor cloud infrastructure, reducing system downtime by 20% and enabling data-driven decision-making across teams.
- Automated ETL workflows using Python and AWS Glue, streamlining data integration processes and cutting integration time by 40%.
- Consolidated and unified multi-source data into a centralized Snowflake data warehouse, improving cross-team data accessibility by 35% and enhancing reporting efficiency.
- Deployed predictive machine learning models in AWS SageMaker, reducing cloud operational costs by 25% through proactive resource optimization.
- Developed SQL-based server health monitoring tools, decreasing unexpected system failures by 30% and improving operational reliability.
- Integrated AWS S3 and RDS with Snowflake, enhancing query performance by 18% and enabling faster data retrieval for analytics.

- Conducted exploratory data analysis (EDA) using Pandas and Matplotlib, identifying cost-saving opportunities and driving a 15% improvement in cloud cost optimization.
- Led end-to-end machine learning pipelines to predict server performance issues, enabling proactive scaling and reducing system failures by

Key Achievements:

- Reduced reporting time by automating SQL scripts and dashboard development.
- Enhanced customer segmentation with ML, increasing marketing efficiency.
- Streamlined analytics and reporting capabilities by migrating data to cloud platforms.

PROJECTS

A Novel Intrusion Classification with Effective Features and Deep Learning Models

Team Members: Divya Sai Kodati, Gopi Krishna Vicharapu, Pavan Venkata Phanisree Peram

Supervising Professor: Dr. Jiangjiang (Jane) Liu

Description:

- Designed and implemented a binary classification model for intrusion detection using Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) models.
- Conducted feature selection using the Chi-Square statistical test, selecting top 10 features to mitigate overfitting and improve model performance
- Evaluated models on benchmark datasets (KDD and BotNetIoT), achieving high accuracy: 96.11% (LSTM on KDD) and 99.08% (DNN on BotNetIoT).
- Demonstrated the effectiveness of deep learning models in detecting anomalous network activity, addressing real-world cybersecurity challenges.
- Feature Engineering: We conducted a thorough analysis of the datasets to identify and select the most relevant features, aiming to improve model performance and reduce computational complexity.

EDUCATION

Masters in Computer Science | Lamar University, Houston, TX – USA

GPA: 3.5/4

Bachelors in Electrical and Electronics Engineering | Vasireddy Venkatadri Institute of Technology, Andhra Pradesh- IND

GPA: 7.6/10

CERTIFICATIONS

Microsoft Power BI Data Analyst Associate (PL-300)

SQL Server Analysis Services (SSAS)

Data Cleaning in Python Essential Training

Introduction to Career Skills in Data Analytics